

Ben²

Bridges

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Project Goal

The goal of our project, **Ben² Bridges**, is to create a decision-support tool that helps governments decide which bridges should receive maintenance when funding is limited. Instead of only prioritizing the oldest bridges or bridges currently in the worst condition, our system will use historical inspection data to predict which bridges are most likely to deteriorate.

The system will combine predicted deterioration with factors such as current condition, traffic volume, detour impact, and estimated maintenance cost. It will then recommend which bridges should be prioritized under a selected budget.

One idea our group considered was **StudySpot**, an AI system that would use computer vision to estimate how crowded campus study spaces are and combine that information with factors such as noise, outlets, lighting, accessibility, and operating hours to recommend suitable locations. However, we realized that collecting enough diverse data would be difficult because of privacy concerns and the need for repeated observations, and the recommendation itself would not truly require a trained AI model since a small number of study spaces could be ranked effectively with simple rules and filters. As a result, AI would feel added on rather than essential to the project. Therefore, we discarded the idea.

Project Scope

Because we only have five to six days, we will develop a focused prototype rather than a complete government infrastructure-management system. We will limit the project to highway bridges in **Pennsylvania** and use publicly available National Bridge Inventory data.

The project will include:

- Cleaning and organizing historical bridge data
- Predicting whether a bridge's condition is likely to decline
- Considering other factors such as traffic volume, detour impact, etc.
- Calculating a risk and priority score for each bridge
- Comparing different maintenance-selection strategies
- Selecting bridges under a hypothetical limited budget
- Displaying the results through an interactive dashboard or map

We will compare BridgeBudget with simpler strategies, including prioritizing the oldest bridges, the busiest bridges, and the bridges currently in the worst condition.

We will not attempt to predict bridge collapse, calculate exact engineering repair costs, or simulate the entire road network. Any repair costs or intervention benefits used in the project will be clearly described as estimates or relative cost units.

How We Will Achieve the Goal Within Our Constraints

We will focus first on creating a minimum viable product containing the most important parts of the project.

On the first day, we will download, inspect, and clean the Pennsylvania bridge datasets. On the second day, we will match bridges across years, create the deterioration target, and train the prediction models. On the third day, we will create the bridge-risk formula, estimated maintenance costs, comparison policies, and budget-allocation system.

On the fourth day, we will develop the interactive dashboard and map. On the fifth day, we will evaluate the model, create graphs, write the methodology and limitations, and prepare the presentation. On the final day, we will integrate all parts of the project, test the dashboard, fix errors, and practice the demonstration.

To save time, we will use established Python libraries such as pandas, scikit-learn, SciPy, Plotly, and Streamlit. We will also precompute the bridge predictions and

allocation results so that the dashboard does not need to retrain the model every time it runs.

Group Member Responsibilities

The responsibilities will overlap so that no major part of the project depends on only one person. Each member will have a primary responsibility while also helping with another area.

Aarav and Aviral: Data Collection, Cleaning, Machine Learning, Model Evaluation

Primary responsibilities:

- Create the variable showing whether a bridge deteriorated
- Select and engineer model features
- Train the logistic regression and tree-based models
- Test the model using a time-based split
- Evaluate calibration, precision, recall, and top-risk bridge identification
- Download the Pennsylvania bridge datasets
- Identify the important columns
- Clean missing or incorrectly formatted values
- Match bridge records across multiple years
- Create the final dataset used by the model

Benjamin Z and Benjamin Y: Risk Scoring, Budget Allocation, Dashboard and Visualizations

Primary responsibilities:

- Develop the maintenance priority score by combining predicted deterioration, current bridge condition, traffic volume, detour impact, and estimated maintenance cost
- Build the budget-allocation system to select which bridges should receive maintenance under a user-selected budget
- Implement and compare alternative selection policies, including oldest bridges, busiest bridges, and bridges in the worst current condition
- Build the Streamlit dashboard
- Create the interactive bridge map

- Add county, budget, and policy controls
- Create graphs comparing the different policies
- Organize the final report, presentation, and demonstration
- Develop the risk and consequence formulas
- Create the estimated maintenance-cost system
- Implement the different comparison policies
- Build the budget-constrained allocation method
- Compare results at several budget levels

Shared Responsibilities

All four group members will participate in:

- Choosing the final project assumptions
- Testing the system
- Reviewing the code and results
- Writing the limitations
- Preparing the presentation
- Practicing the final demonstration

This overlap will allow another group member to continue a task if someone becomes stuck or unavailable.

Backup Plan

If we have difficulty matching bridges across many years, we will reduce the number of years and use only datasets with consistent identifiers and formats.

If there are too few deterioration examples, we will define deterioration as a decline in any major component, deck, superstructure, or substructure, instead of only a decline in the bridge's minimum condition rating.

If the advanced machine-learning model does not work well, we will use logistic regression or a simpler rule-based deterioration score. The project can still compare predictive prioritization with condition-, age-, and traffic-based strategies.

If the optimization method becomes too complicated, we will use a benefit-per-cost ranking. Bridges will be sorted by expected benefit divided by estimated cost and selected until the available budget is exhausted.

If the interactive map does not work, we will use a simpler table, static map, and policy-comparison graphs. If online deployment fails, we will run the Streamlit dashboard locally and prepare screenshots or a recorded demonstration as a backup.

If one group member becomes stuck, another member working on an overlapping task will help troubleshoot or temporarily take over that part. We will save shared code and data in a common repository so that every member can access the latest version.

Our priority will be to complete a reliable minimum viable product containing cleaned data, one working prediction method, several comparison strategies, a budget-allocation method, and a clear visual presentation. Optional features will only be added after all core components are working.